INCH-POUND

MIL-I-85071/3B(AS)
16 September 1991

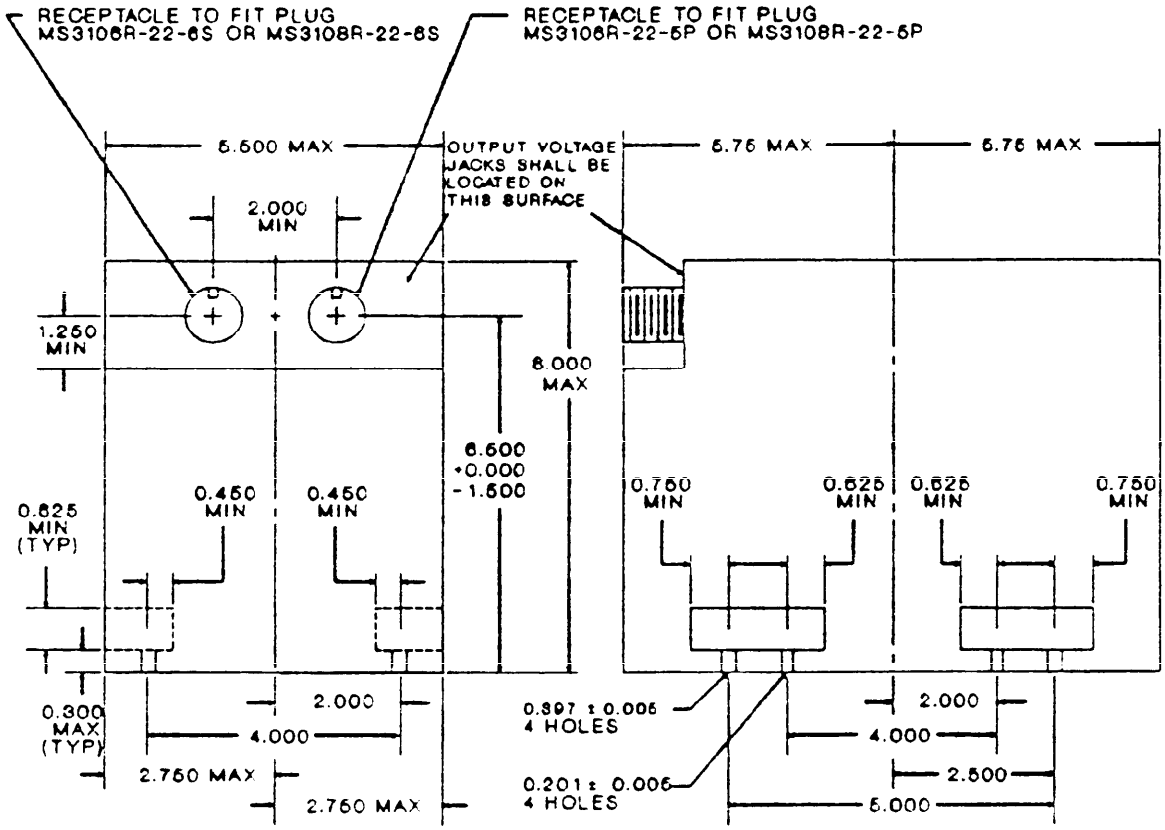
SUPERSEDING
MIL-I-85071/3A(AS)
28 FEBRUARY 1985

MILITARY SPECIFICATION SHEET

INVERTER, AIRCRAFT, DC TO AC, STATIC, 750 VA 3-PHASE, 1000 VA 1-PHASE, AND 1000 VA 3-PHASE, MULTIPURPOSE, CLASS B

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-85071.

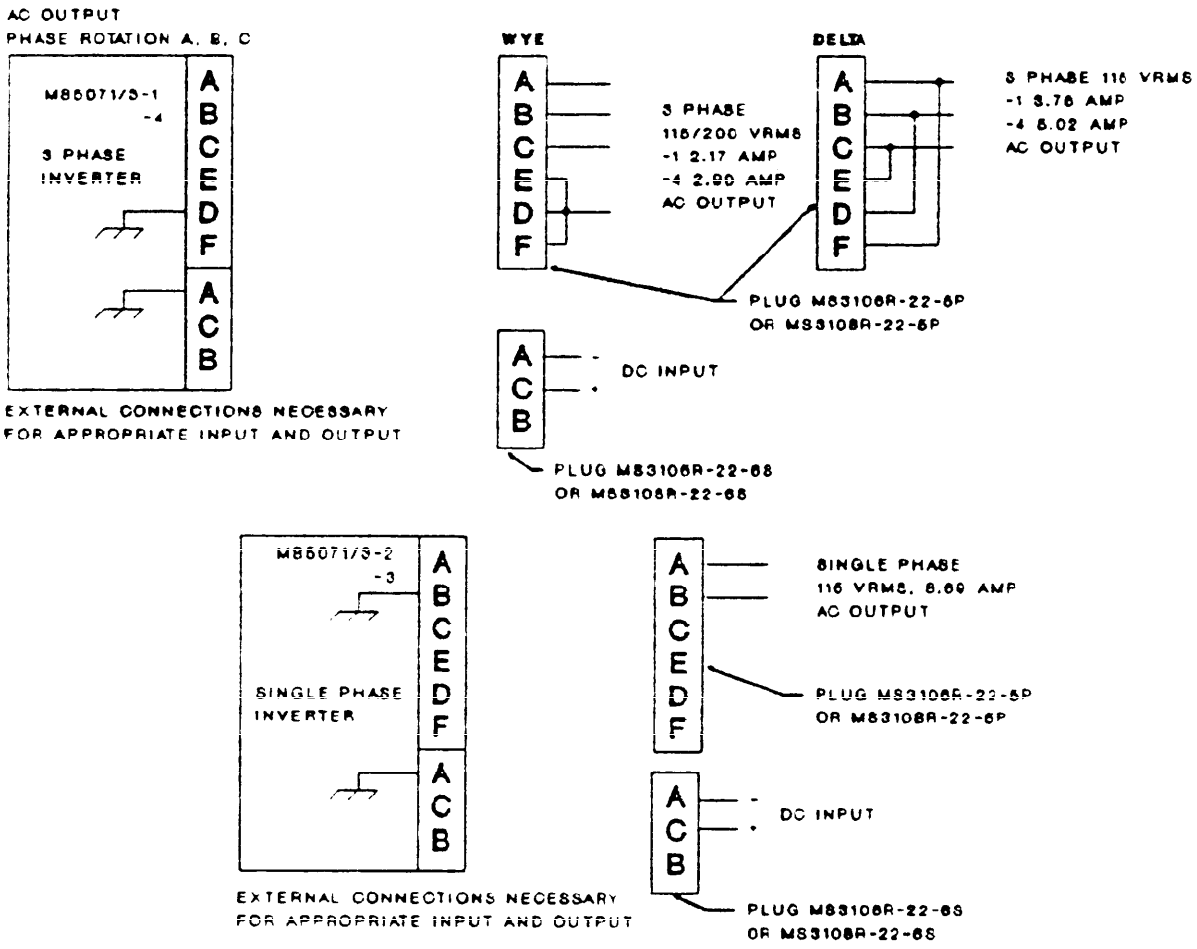


* Airflow exhaust from the connector end for the -2 and -3 inverters
Airflow exhaust from the end opposite the connectors for the -1 and -4 inverters

AMSC N/A FSC 6130

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DESCRIPTION

PART NO.	POWER RATING	POWER FACTOR	MAX WEIGHT (POUNDS)	SUPERSEDING
M85071/3-1	750 VA, 3ϕ	0.75 MIN LAG 0.95 MIN LEAD	23	MS17406-3
M85071/3-2	1000 VA, 1ϕ		23	MS17406-4
M85071/3-3	1000 VA, 1ϕ LOW INPUT VOLTAGE		23	MS17406-5
M85071/3-4	1000 VA, 3ϕ		23	

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REQUIREMENTS:

1. The inverter shall operate under the following conditions:

For any combination of conditions listed within each section below, the output voltage and frequency shall remain within the indicated limits.	OUTPUT LIMITS LIMITS APPLY TO EACH PHASE		
	RMS VOLTAGE	FREQ (Hz)	NOTES
A. NORMAL INPUT VOLTAGE 1. Input voltage varied between 26 and 29 volts 2. Load varied from zero to full load 3. Ambient temperature-altitude varied per Class B 4. Load power factor varied from .75 lag to .95 lead 5. Unit mounted in any position	Low 112.5 High 117.5	Low 393 High 407	
B. ABNORMAL INPUT VOLTAGE 6. Variations 2, 3, 4, and 5 above with input voltage varied between 19.5 and 26 volts and load from zero to rated full load conditions	Low 108 High 117.5	Low 393 High 407	
C. OVERLOAD 7. Variations 1 through 5 above except load varied from rated full load current to 200% rated full load current for 5 seconds	Low 50 High 117.5	Low 393 High 407	
D. LOW INPUT VOLTAGE 8. Input voltage varied between 10 and 19.5 volts for 3 minutes * 9. Load impedance varied from infinity to 26.5 Ohms for the -2 inverter, from infinity to 106 Ohms/phase for the -1 inverter, and from infinity to 79.4 Ohms/phase for the -4 inverter 10. Ambient temperature varied between -35°C and +40°C at sea level with inverter fully warmed up, and between -35°C and +85°C at sea level with inverter not previously warmed up 11. Variations 4 and 5 above	Low 45 High 117.5	Low 393 High 407	Applicable to -1, -2, and -4 Inverter
12. Variations 8, 9, 10, and 11 above	Low 108 High 117.5	Low 393 High 407	Applicable to -3 Inverter only

2. At altitudes above 50,000 feet, the -2 and -3 inverters are rated at 750 VA.

* 3. The -1 and -4 inverters shall operate either in the wye or delta mode when external connections are made as shown in the connection diagram.

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THE FOLLOWING REQUIREMENTS APPLY TO THE -2 INVERTER ONLY:

4. Add the following to paragraph 3.7.1.6: except the minimum limit shall be 80 peak Volts/$\sqrt{2}$ from 0 to 0.01 seconds instead of 84 peak Volts/$\sqrt{2}$.

5. During the ripple susceptibility test of paragraph 4.7.2.12.3 the output voltage modulation shall not exceed 1.2%.

6. Inverter samples 3 and 4 shall be tested for conformance to the low input voltage requirements in section D.

THE FOLLOWING REQUIREMENTS APPLY TO THE -3 INVERTER ONLY:

7. The -3 inverter shall meet the requirements of the -2 inverter in addition to the -3 low input voltage operating requirements.

8. The total harmonic and extraneous content shall not exceed 8% of the fundamental under any rated operating condition with input voltage between 10 and 19.5 Volts.

9. The inverter efficiency shall be 50% minimum at input voltages between 10 and 19.5 Volts, sea level altitude, 20°C ± 15°C ambient temperature, 26.5 Ohm load, and unity power factor.

10. The inverter shall be subjected to 1250 consecutive cycles of operation consisting of 1 minute at 10 Volts input and 26.5 Ohm load, and 4 minutes at 28 Volts input and 1000 VA load. This test shall be conducted at room temperature with unity power factor loads. At the conclusion of the test the inverter shall be tested for compliance with requirements for voltage and frequency regulation at room ambient, paragraph 4.7.2.2.1. For qualification this test shall be conducted on the number 4 sample.

NOTES:

1. All dimensions are in inches.

2. Unless otherwise specified, tolerances are ±.015 inches.

3. The margins of this specification sheet are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing Activity

NAVY - AS

Project No. 6130-N334

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